Paper List

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Abstract

整理一系列文章为了后续能快速的找到内容.

1 World Model

1.1 JEPAs

Training object is (likely) representations, and use CEM to MPC.

- I-JEPA,V-JEPA [1, 2, 3, 4, 5]
- leWM-based [6, 7, 8],
- DINO-WM [9]
- [10], Hierarchical planning

1.2 latent representations

I think that JEPA also helps for building a 'good' representations of model prediction. Here are works that focusing on training the encoder instead of just next-step prediction.

- [11], smoothness for representations
- [12], project DINO token
- *[13], planning in latent*
- [14],scaling LeWM

1.3 WAMs

- [15], predict future
- [16], use optical-flow. *TODO:add lfy's work*
- [17], just denoising w/o predict future
- [18], statistical work to analyze ACTION SPACE
- [19, 20], atom ops of action

1.4 Others

- [21],Point WMs
- [22], dataset for other goals
- [23], testing on Video-gen
- [12], use ODE as training objective
- [24], good pipeline on WM
- [25, 26], ARC-AGI (idk where to put it)
- [27], Probe to find whether WM can find the original laws(e.g.,gravity)

2 Loop

- [28],LoopViT
- [29], use DP to find training-free loops in ViTs
- [30], refine in Diffusion
- [31], LoopDiT
- [32], LoopWM

3 NCA

- [33], Summary
- [34], Diffusion as NCA
- [35, 36], 2D reasoning w/ NCA
- Yitao Xu[1]'s prior work:
  - Growing NCA [37], the original NCA
  - DyNCA [38], dynamic texture synthesis by motion supervision
  - MeshNCA [39], texture synthesis on meshes
  - [40], NCA + appearance decoder, from cells to high-res pixels
  - NoiseNCA [41], noisy seed for spatio-temporal continuity
  - [42], Neural Particle Automata

4 Theory

- [43],question ID/OOD acc using **gradient flow**. Check if u wish to see analyses of subspace
- [44],great work of "capacity" of models[2]
- several works of JEPAs.
  - [7],Gaussian world, how SIGReg work ?
  - [45], LpWM , sprase-space analysis
  - [11], smoothness for representations
  - [46], decode state of WMs to identify
- [47], random matrixs
- [48], Identity in ResNet.Tengyu Ma play
- [49],low-rank regression

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[1] thx for his wonderful intuition!
[2] here to say, inductive bias is needed for models to learn

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